FACULTY SEMINAR · CALCULUS

Section 4.2 · Applications of Differentiation

Concavity and Inflection Points

$$f''(x) > 0 \Rightarrow \smile \qquad f''(x) < 0 \Rightarrow \frown$$

Dr. Wessam Gouda

Department of Mathematics

Agenda

01 Learning Objectives

02 Concavity: definition & test

03 Interactive: Concavity Explorer

04 Related Problem 1

05 Inflection points

06 Example 2

07 Examples 3, 6, 8, 9

08 The Second Derivative Test

09 Example 10 (1)

10 Challenging Problems 5

11 Communicate with Mathematics 2

Learning Objectives

At the end of this section, you should be able to:

- Investigate additional features of graphs of functions, particularly concavity and points of inflection.

Learning Objectives

At the end of this section, you should be able to:

- Investigate additional features of graphs of functions, particularly concavity and points of inflection.
- Explore and analyze the concavity of graphs of functions using the methods of calculus.

Learning Objectives

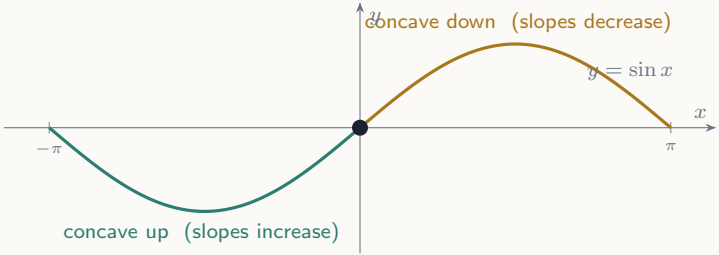
At the end of this section, you should be able to:

- Investigate additional features of graphs of functions, particularly concavity and points of inflection.
- Explore and analyze the concavity of graphs of functions using the methods of calculus.
- Identify and determine points of inflection of a function using calculus techniques.

Concavity

Interactive ↗

The first derivative is the slope of the tangent line; the **second derivative** describes how that slope changes — the *curvature* of the graph.



Concave up: the graph lies **above** its tangent lines.

Concave down: the graph lies **below** its tangent lines.

Definition 4.2.1 & Theorem 4.2.1

Definition 4.2.1 (Concave Up or Concave Down)

Let f be a differentiable function on an open interval I. Then

1. f is said to be **concave up** on I if f' is increasing on I.
2. f is said to be **concave down** on I if f' is decreasing on I.

Definition 4.2.1 & Theorem 4.2.1

Definition 4.2.1 (Concave Up or Concave Down)

Let f be a differentiable function on an open interval I. Then

1. f is said to be **concave up** on I if f' is increasing on I.
2. f is said to be **concave down** on I if f' is decreasing on I.

Theorem 4.2.1 (Concavity Test)

Suppose $f''(x)$ exists on an open interval I.

1. If $f''(x) > 0$ on I, then f is concave up on I.
2. If $f''(x) < 0$ on I, then f is concave down on I.

Interactive: Concavity Explorer

Drag a point along the curve and watch:

- the tangent line and its slope $f'(x)$,
- the sign of $f''(x)$,
- the curve coloured concave up / concave down,
- the inflection points marked.

Functions: the examples of this seminar.

Open the Concavity Explorer

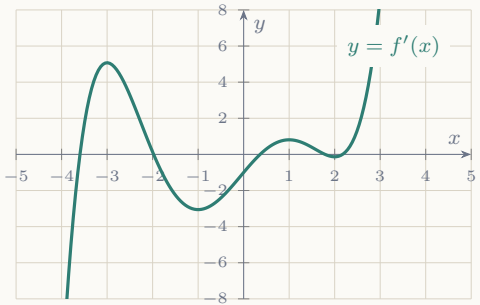
https://alostazy2k.github.io/seminars/4.2/concavity-explorer.html

offline copy (same folder as this PDF)

Related Problem 1

Interactive ↗

Use the graph of $y = f'(x)$, given in the figure, to find the following: 1. The intervals on which f is concave up. 2. The intervals on which f is concave down.



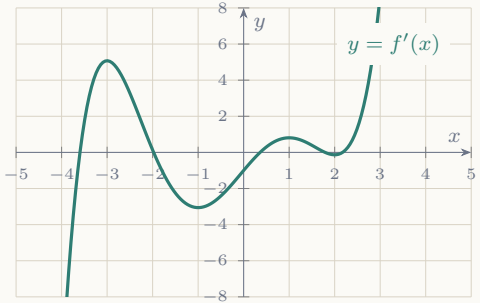
graph redrawn with the same turning points as Figure 4.2.4

Answer

Related Problem 1

Interactive ↗

Use the graph of $y = f'(x)$, given in the figure, to find the following: 1. The intervals on which f is concave up. 2. The intervals on which f is concave down.



graph redrawn with the same turning points as Figure 4.2.4

Answer

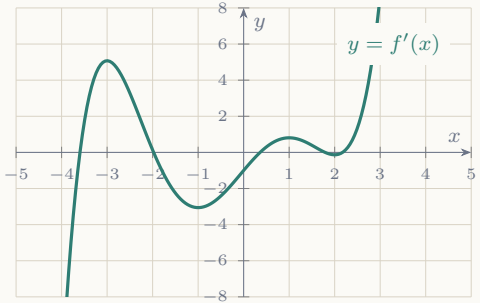
f' increasing on
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

$\Rightarrow$ **1. Concave up on**
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

Related Problem 1

Interactive ↗

Use the graph of $y = f'(x)$, given in the figure, to find the following: 1. The intervals on which f is concave up. 2. The intervals on which f is concave down.



graph redrawn with the same turning points as Figure 4.2.4

Answer

f' increasing on
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

$\Rightarrow$ **1. Concave up on**
$(-\infty, -3)$, $(-1, 1)$, $(2, \infty)$

f' decreasing on $(-3, -1)$, $(1, 2)$

$\Rightarrow$ **2. Concave down on** $(-3, -1)$, $(1, 2)$

Definition 4.2.2 & Theorem 4.2.2

Definition 4.2.2 (Inflection Point)

Let f be a continuous function on an open interval I and $x_0 \in I$. Then the point $(x_0, f(x_0))$ is said to be an **inflection point** if $f(x)$ changes its concavity (upward to downward or downward to upward) around x_0.

Definition 4.2.2 & Theorem 4.2.2

Definition 4.2.2 (Inflection Point)

Let f be a continuous function on an open interval I and $x_0 \in I$. Then the point $(x_0, f(x_0))$ is said to be an **inflection point** if $f(x)$ changes its concavity (upward to downward or downward to upward) around x_0.

Theorem 4.2.2 (Points of Inflection)

Let $(c, f(c))$ be a point of inflection of the graph of the function f. Then, either $f''(c) = 0$, or $f''(c)$ does not exist.

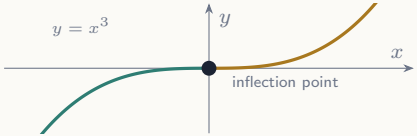
Definition 4.2.2 & Theorem 4.2.2

Definition 4.2.2 (Inflection Point)

Let f be a continuous function on an open interval I and $x_0 \in I$. Then the point $(x_0, f(x_0))$ is said to be an **inflection point** if $f(x)$ changes its concavity (upward to downward or downward to upward) around x_0.

Theorem 4.2.2 (Points of Inflection)

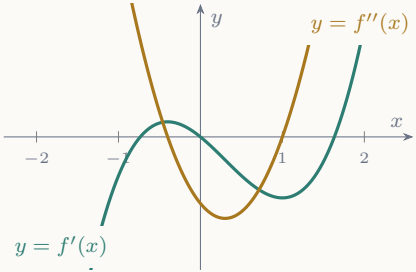
Let $(c, f(c))$ be a point of inflection of the graph of the function f. Then, either $f''(c) = 0$, or $f''(c)$ does not exist.



Example 2

Interactive ↗

Consider the graphs of $y = f'(x)$ and $y = f''(x)$. Determine: 1. the intervals on which f is concave up; 2. the intervals on which f is concave down; 3. the x-coordinates of all inflection points of $f(x)$ (if any).



graphs redrawn with the same zeros of f'' as Figure 4.2.7

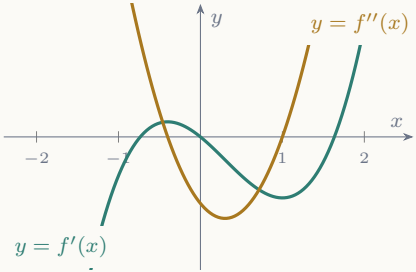
Solution

Using the graphs and the test for concavity:

Example 2

Interactive ↗

Consider the graphs of $y = f'(x)$ and $y = f''(x)$. Determine: 1. the intervals on which f is concave up; 2. the intervals on which f is concave down; 3. the x-coordinates of all inflection points of $f(x)$ (if any).



graphs redrawn with the same zeros of f'' as Figure 4.2.7

Solution

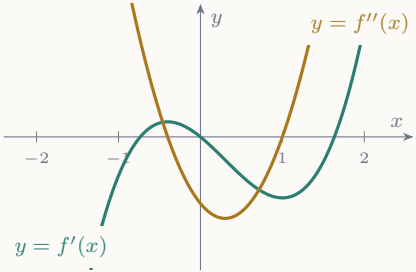
Using the graphs and the test for concavity:

1. f is concave up on $\left(-\infty, -\frac{2}{5}\right),\ (1, \infty)$

Example 2

Interactive ↗

Consider the graphs of $y = f'(x)$ and $y = f''(x)$. Determine: 1. the intervals on which f is concave up; 2. the intervals on which f is concave down; 3. the x-coordinates of all inflection points of $f(x)$ (if any).



graphs redrawn with the same zeros of f'' as Figure 4.2.7

Solution

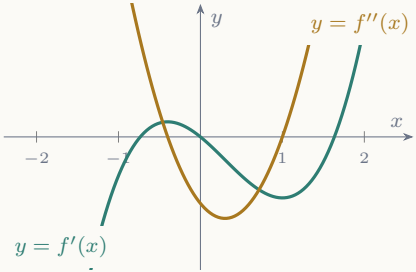
Using the graphs and the test for concavity:

1. f is concave up on $\left(-\infty, -\frac{2}{5}\right),\ (1, \infty)$
2. f is concave down on $\left(-\frac{2}{5}, 1\right)$

Example 2

Interactive ↗

Consider the graphs of $y = f'(x)$ and $y = f''(x)$. Determine: 1. the intervals on which f is concave up; 2. the intervals on which f is concave down; 3. the x-coordinates of all inflection points of $f(x)$ (if any).



graphs redrawn with the same zeros of f'' as Figure 4.2.7

Solution

Using the graphs and the test for concavity:

1. f is concave up on $\left(-\infty, -\frac{2}{5}\right),\ (1, \infty)$
2. f is concave down on $\left(-\frac{2}{5}, 1\right)$
3. Inflection points of f occur at $\boxed{x = -\frac{2}{5} \text{ and } x = 1}$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution f is a polynomial: defined for all $x \in \mathbb{R}$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$$f'(x) = 3x^2 - 12x + 9$$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$f'(x) = 3x^2 - 12x + 9$

$f''(x) = 6x - 12$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

$$6x - 12 = 0 \implies x = 2$$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

$$6x - 12 = 0 \implies x = 2$$

Intervals: $(-\infty, 2),\ (2, \infty)$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

$$6x - 12 = 0 \implies x = 2$$

Intervals: $(-\infty, 2),\ (2, \infty)$

Interval	$(-\infty, 2)$	$(2, \infty)$
Test value k	0	3
Sign of $f''(k)$	$-$	$+$
Behavior of $f(x)$	concave downward	concave upward

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

$$6x - 12 = 0 \implies x = 2$$

Intervals: $(-\infty, 2),\ (2, \infty)$

Interval	$(-\infty, 2)$	$(2, \infty)$
Test value k	0	3
Sign of $f''(k)$	$-$	$+$
Behavior of $f(x)$	concave downward	concave upward

Concave up on $(2, \infty)$, concave down on $(-\infty, 2)$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

$$6x - 12 = 0 \implies x = 2$$

Intervals: $(-\infty, 2)$, $(2, \infty)$

Interval	$(-\infty, 2)$	$(2, \infty)$
Test value k	0	3
Sign of $f''(k)$	$-$	$+$
Behavior of $f(x)$	concave downward	concave upward

Concave up on $(2, \infty)$, concave down on $(-\infty, 2)$

$f(2) = 32 \implies$ $\boxed{(2,\ 32) \text{ is an inflection point}}$

Example 3

Interactive ↗

For $f(x) = x^3 - 6x^2 + 9x + 30$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a polynomial: defined for all $x \in \mathbb{R}$

$$f'(x) = 3x^2 - 12x + 9$$

$$f''(x) = 6x - 12$$

$$6x - 12 = 0 \implies x = 2$$

Intervals: $(-\infty, 2),\ (2, \infty)$

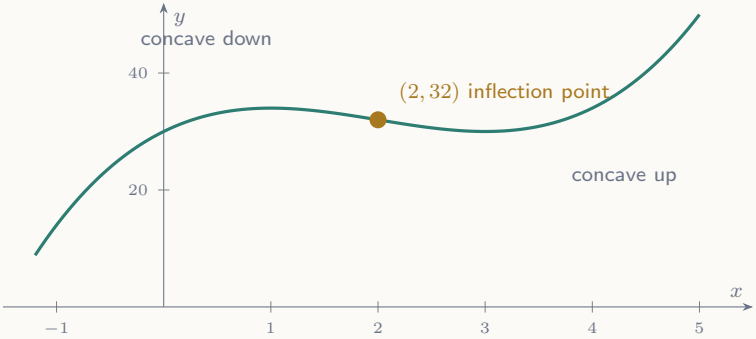
Interval	$(-\infty, 2)$	$(2, \infty)$
Test value k	0	3
Sign of $f''(k)$	$-$	$+$
Behavior of $f(x)$	concave downward	concave upward

Concave up on $(2, \infty)$, concave down on $(-\infty, 2)$

$f(2) = 32 \implies$ $\boxed{(2,\ 32) \text{ is an inflection point}}$

Example 3 — graph

Interactive ↗



Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

$$f''(x) = \frac{16}{(x-5)^3}$$

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

$$f''(x) = \frac{16}{(x-5)^3}$$

$f''(x) \neq 0$, and $(x-5)^3 = 0 \Rightarrow x = 5$: $f''(x)$ does not exist at $x = 5$

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

$$f''(x) = \frac{16}{(x-5)^3}$$

$f''(x) \neq 0$, and $(x-5)^3 = 0 \Rightarrow x = 5$: $f''(x)$ does not exist at $x = 5$

x	$-\infty$		5		$+\infty$
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

$$f''(x) = \frac{16}{(x-5)^3}$$

$f''(x) \neq 0$, and $(x-5)^3 = 0 \Rightarrow x = 5$: $f''(x)$ does not exist at $x = 5$

x	$-\infty$		5		$+\infty$
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Concave up on $(5, \infty)$, concave down on $(-\infty, 5)$

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

$$f''(x) = \frac{16}{(x-5)^3}$$

$f''(x) \neq 0$, and $(x-5)^3 = 0 \Rightarrow x = 5$: $f''(x)$ does not exist at $x = 5$

x	$-\infty$		5		$+\infty$
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Concave up on $(5, \infty)$, concave down on $(-\infty, 5)$

No inflection points, although the concavity changes around $x = 5$,

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

$$f''(x) = \frac{16}{(x-5)^3}$$

$f''(x) \neq 0$, and $(x-5)^3 = 0 \Rightarrow x = 5$: $f''(x)$ does not exist at $x = 5$

x	$-\infty$		5		$+\infty$
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Concave up on $(5, \infty)$, concave down on $(-\infty, 5)$

No inflection points, although the concavity changes around $x = 5$,

because $5 \notin D_f$

Example 6

Interactive ↗

For $f(x) = \dfrac{x+3}{x-5}$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is rational: $D_f = (-\infty, 5) \cup (5, \infty)$, twice differentiable at any $x \neq 5$

$$f'(x) = \frac{-8}{(x-5)^2}$$

$$f''(x) = \frac{16}{(x-5)^3}$$

$f''(x) \neq 0$, and $(x-5)^3 = 0 \Rightarrow x = 5$: $f''(x)$ does not exist at $x = 5$

x	$-\infty$		5		$+\infty$
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

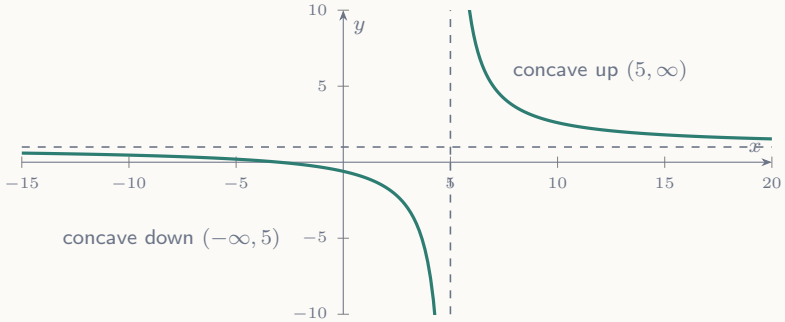
Concave up on $(5, \infty)$, concave down on $(-\infty, 5)$

No inflection points, although the concavity changes around $x = 5$,

because $5 \notin D_f$

Example 6 — graph

Interactive ↗



Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

$f''(x)$ does not exist at $x = 0$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

Concave up on $\left(-\infty, -\frac{1}{5}\right)$; concave down on $\left(-\frac{1}{5}, 0\right)$, $(0, \infty)$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$-2(5x+1) = 0 \implies x = -\frac{1}{5} \implies f''\left(-\frac{1}{5}\right) = 0$

$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

Concave up on $\left(-\infty, -\frac{1}{5}\right)$; concave down on $\left(-\frac{1}{5}, 0\right)$, $(0, \infty)$

f'' does not change sign around $x = 0$: no inflection at $x = 0$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

Concave up on $\left(-\infty, -\frac{1}{5}\right)$; concave down on $\left(-\frac{1}{5}, 0\right), (0, \infty)$

f'' does not change sign around $x = 0$: no inflection at $x = 0$

f'' changes sign around $x = -\frac{1}{5}$ (positive to negative):

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

Concave up on $\left(-\infty, -\frac{1}{5}\right)$; concave down on $\left(-\frac{1}{5}, 0\right)$, $(0, \infty)$

f'' does not change sign around $x = 0$: no inflection at $x = 0$

f'' changes sign around $x = -\frac{1}{5}$ (positive to negative):

only inflection point $= \boxed{\left(-\frac{1}{5},\ \frac{6}{5}\sqrt[3]{\frac{1}{25}}\right)}$

Example 8

Interactive ↗

For $f(x) = x^{2/3}(1-x)$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is a radical function: $D_f = (-\infty, \infty)$, twice differentiable at $x \neq 0$

$$f'(x) = \frac{2-5x}{3x^{1/3}}$$

$$f''(x) = \frac{-2(5x+1)}{9x^{4/3}}$$

$$-2(5x+1) = 0 \implies x = -\tfrac{1}{5} \implies f''\left(-\tfrac{1}{5}\right) = 0$$

$f''(x)$ does not exist at $x = 0$

x	$-\infty$		$-\frac{1}{5}$		0		$+\infty$
$f''(x)$		$+$		$-$		$-$	
$f(x)$		$\smile$		$\frown$		$\frown$	

Concave up on $\left(-\infty, -\frac{1}{5}\right)$; concave down on $\left(-\frac{1}{5}, 0\right)$, $(0, \infty)$

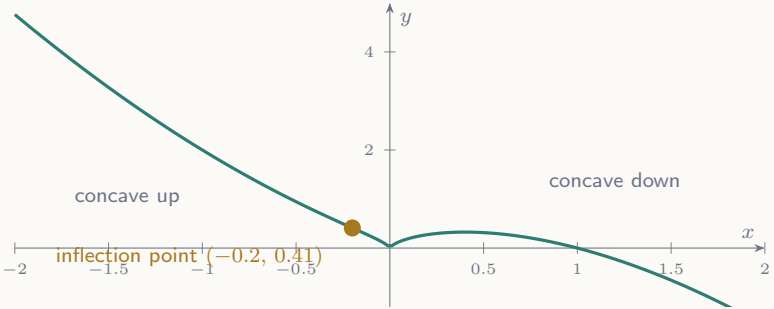
f'' does not change sign around $x = 0$: no inflection at $x = 0$

f'' changes sign around $x = -\frac{1}{5}$ (positive to negative):

only inflection point $= \boxed{\left(-\frac{1}{5},\ \frac{6}{5}\sqrt[3]{\frac{1}{25}}\right)}$

Example 8 — graph

Interactive ↗



Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution f is continuous and twice differentiable on $[0, 2\pi]$

Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is continuous and twice differentiable on $[0, 2\pi]$

$f'(x) = 1 + \cos x$

Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is continuous and twice differentiable on $[0, 2\pi]$

$$f'(x) = 1 + \cos x$$

$$f''(x) = -\sin x$$

Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is continuous and twice differentiable on $[0, 2\pi]$

$$f'(x) = 1 + \cos x$$

$$f''(x) = -\sin x$$

$$f''(x) = 0 \implies x = \pi$$

Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is continuous and twice differentiable on $[0, 2\pi]$

$f'(x) = 1 + \cos x$

$f''(x) = -\sin x$

$f''(x) = 0 \implies x = \pi$

x	0		π		2π
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is continuous and twice differentiable on $[0, 2\pi]$

$f'(x) = 1 + \cos x$

$f''(x) = -\sin x$

$f''(x) = 0 \implies x = \pi$

x	0		π		2π
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Concave up on $(\pi, 2\pi)$, concave down on $(0, \pi)$

Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is continuous and twice differentiable on $[0, 2\pi]$

$f'(x) = 1 + \cos x$

$f''(x) = -\sin x$

$f''(x) = 0 \implies x = \pi$

x	0		π		2π
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Concave up on $(\pi, 2\pi)$, concave down on $(0, \pi)$

$(\pi, f(\pi)) = \boxed{(\pi,\ \pi)}$ is the only inflection point

Example 9

Interactive ↗

For $f(x) = x + \sin x,\ x \in [0, 2\pi]$, find the intervals on which f is concave up or down, and find all inflection points (if any).

Solution

f is continuous and twice differentiable on $[0, 2\pi]$

$f'(x) = 1 + \cos x$

$f''(x) = -\sin x$

$f''(x) = 0 \implies x = \pi$

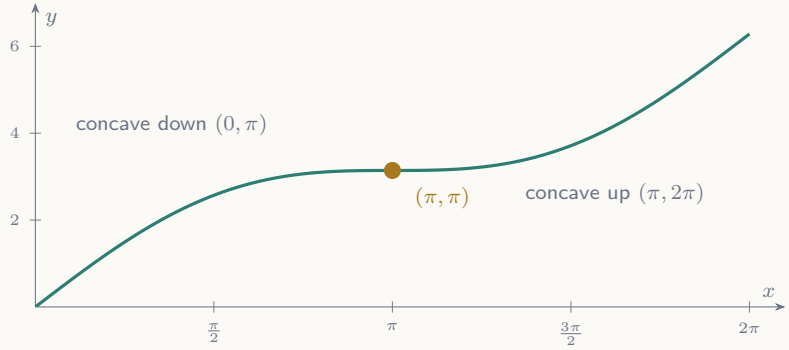
x	0		π		2π
$f''(x)$		$-$		$+$	
$f(x)$		$\frown$		$\smile$	

Concave up on $(\pi, 2\pi)$, concave down on $(0, \pi)$

$(\pi, f(\pi)) = \boxed{(\pi,\ \pi)}$ is the only inflection point

Example 9 — graph

Interactive ↗



Theorem 4.2.3 (The Second Derivative Test)

Theorem 4.2.3

Suppose f'' is continuous on an open interval containing c with $f'(c) = 0$.

- If $f''(c) > 0$, then f has a relative minimum at c.
- If $f''(c) < 0$, then f has a relative maximum at c.
- If $f''(c) = 0$, then the test is inconclusive; f may have a relative maximum, relative minimum, or neither at c.

In such cases ($f''(c) = 0$), we may use the first derivative test.

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

$$12(x+1)(x-1)^2 = 0 \implies x = -1,\ x = 1$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

$$12(x+1)(x-1)^2 = 0 \implies x = -1,\ x = 1$$

$$f''(-1) = 48 > 0, \quad f''(1) = 0$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

$$12(x+1)(x-1)^2 = 0 \implies x = -1,\ x = 1$$

$$f''(-1) = 48 > 0, \quad f''(1) = 0$$

By the Second Derivative Test: **relative minimum at** $x = -1$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

$$12(x+1)(x-1)^2 = 0 \implies x = -1,\ x = 1$$

$$f''(-1) = 48 > 0, \quad f''(1) = 0$$

By the Second Derivative Test: **relative minimum at** $x = -1$

At $x = 1$: test inconclusive; f' does not change sign at $x = 1$:

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

$$12(x+1)(x-1)^2 = 0 \implies x = -1,\ x = 1$$

$$f''(-1) = 48 > 0, \quad f''(1) = 0$$

By the Second Derivative Test: **relative minimum at** $x = -1$

At $x = 1$: test inconclusive; f' does not change sign at $x = 1$:

$$f'(0.5) = 4.5 > 0, \quad f'(1.5) = \tfrac{15}{2} > 0$$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

$$12(x+1)(x-1)^2 = 0 \implies x = -1,\ x = 1$$

$$f''(-1) = 48 > 0, \quad f''(1) = 0$$

By the Second Derivative Test: **relative minimum at** $x = -1$

At $x = 1$: test inconclusive; f' does not change sign at $x = 1$:

$$f'(0.5) = 4.5 > 0, \quad f'(1.5) = \tfrac{15}{2} > 0$$

$\implies$ no relative maximum or minimum at $x = 1$

Example 10 (1) — relative extrema

Interactive ↗

Find the relative extrema of f using the second derivative test if applicable. Discuss the concavity of f and find the point(s) of inflection: $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2]$.

Solution

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 36x^2 - 24x - 12$$

$$\implies f''(x) = 12(3x^2 - 2x - 1)$$

$$12x^3 - 12x^2 - 12x + 12 = 0$$

$$12(x^3 - x^2 - x + 1) = 0$$

$$12\left[x^2(x-1) - (x-1)\right] = 0$$

$$12(x-1)(x^2-1) = 0$$

$$12(x+1)(x-1)^2 = 0 \implies x = -1,\ x = 1$$

$$f''(-1) = 48 > 0, \quad f''(1) = 0$$

By the Second Derivative Test: **relative minimum at** $x = -1$

At $x = 1$: test inconclusive; f' does not change sign at $x = 1$:

$$f'(0.5) = 4.5 > 0, \quad f'(1.5) = \tfrac{15}{2} > 0$$

$\implies$ no relative maximum or minimum at $x = 1$

Example 10 (1) — concavity

Interactive ↗

$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

Solution

$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

Example 10 (1) — concavity

Interactive ↗

$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

Solution

$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

$(3x + 1)(x - 1) = 0 \implies x = -\frac{1}{3}$ or $x = 1$

Example 10 (1) — concavity

Interactive ↗

$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

Solution

$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

$(3x + 1)(x - 1) = 0 \implies x = -\frac{1}{3}$ or $x = 1$

Intervals: $\left(-2, -\frac{1}{3}\right),\ \left(-\frac{1}{3}, 1\right),\ (1, 2)$

Example 10 (1) — concavity

Interactive ↗

$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

Solution

$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

$(3x + 1)(x - 1) = 0 \implies x = -\frac{1}{3}$ or $x = 1$

Intervals: $\left(-2, -\frac{1}{3}\right),\ \left(-\frac{1}{3}, 1\right),\ (1, 2)$

Interval	$\left(-2, -\frac{1}{3}\right)$	$\left(-\frac{1}{3}, 1\right)$	$(1, 2)$
Test value k	-1	0	1.5
Sign of $f''(k)$	$+$	$-$	$+$
Behavior of $f(x)$	concave upward	concave downward	concave upward

Example 10 (1) — concavity

Interactive ↗

$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

Solution

$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

$(3x + 1)(x - 1) = 0 \implies x = -\frac{1}{3}$ or $x = 1$

Intervals: $\left(-2, -\frac{1}{3}\right),\ \left(-\frac{1}{3}, 1\right),\ (1, 2)$

Interval	$\left(-2, -\frac{1}{3}\right)$	$\left(-\frac{1}{3}, 1\right)$	$(1, 2)$
Test value k	-1	0	1.5
Sign of $f''(k)$	$+$	$-$	$+$
Behavior of $f(x)$	concave upward	concave downward	concave upward

Concavity changes at $x = -\frac{1}{3}$ and $x = 1$:

Example 10 (1) — concavity

Interactive ↗

$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

Solution

$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

$(3x + 1)(x - 1) = 0 \implies x = -\frac{1}{3}$ or $x = 1$

Intervals: $\left(-2, -\frac{1}{3}\right),\ \left(-\frac{1}{3}, 1\right),\ (1, 2)$

Interval	$\left(-2, -\frac{1}{3}\right)$	$\left(-\frac{1}{3}, 1\right)$	$(1, 2)$
Test value k	-1	0	1.5
Sign of $f''(k)$	$+$	$-$	$+$
Behavior of $f(x)$	concave upward	concave downward	concave upward

Concavity changes at $x = -\frac{1}{3}$ and $x = 1$:

inflection points $= \boxed{\left(-\frac{1}{3},\ -\frac{94}{27}\right)}, \boxed{(1,\ 6)}$

Example 10 (1) — concavity

Interactive ↗

$f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1,\ [-2, 2], \quad f''(x) = 12(3x^2 - 2x - 1).$

Solution

$f''(x) = 0$ if $3x^2 - 2x - 1 = 0$

$(3x + 1)(x - 1) = 0 \implies x = -\frac{1}{3}$ or $x = 1$

Intervals: $\left(-2, -\frac{1}{3}\right),\ \left(-\frac{1}{3}, 1\right),\ (1, 2)$

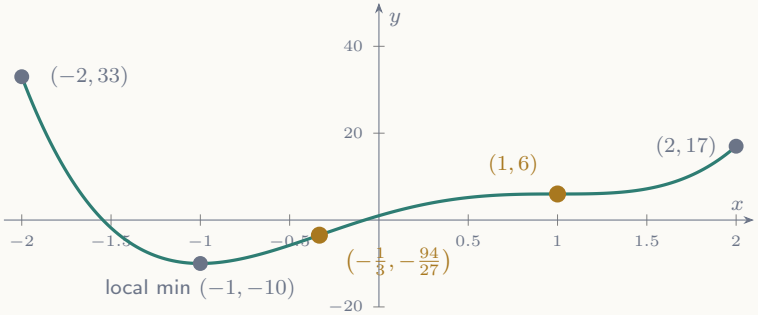
Interval	$\left(-2, -\frac{1}{3}\right)$	$\left(-\frac{1}{3}, 1\right)$	$(1, 2)$
Test value k	-1	0	1.5
Sign of $f''(k)$	$+$	$-$	$+$
Behavior of $f(x)$	concave upward	concave downward	concave upward

Concavity changes at $x = -\frac{1}{3}$ and $x = 1$:

inflection points $= \boxed{\left(-\frac{1}{3},\ -\frac{94}{27}\right)},\ \boxed{(1,\ 6)}$

Example 10 (1) — graph

Interactive ↗



Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution (Assume f, g twice differentiable.)

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \Longrightarrow$ first term ≥ 0; $\quad g'' > 0$

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \implies$ first term ≥ 0; $\; g'' > 0$

$\implies$ $h''(x) > 0$ whenever $f'(g(x)) > 0$

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \implies$ first term ≥ 0; $g'' > 0$

$\implies h''(x) > 0$ whenever $f'(g(x)) > 0$

Condition: f increasing $(f' > 0) \implies h = f \circ g$ is concave up

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \implies$ first term ≥ 0; $g'' > 0$

$\implies h''(x) > 0$ whenever $f'(g(x)) > 0$

Condition: f increasing ($f' > 0$) $\implies$ $h = f \circ g$ is concave up

Without it, it may fail: $f(u) = e^{-u}$ (concave up, decreasing),

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \implies$ first term ≥ 0; $g'' > 0$

$\implies h''(x) > 0$ whenever $f'(g(x)) > 0$

Condition: f increasing ($f' > 0$) $\implies$ $h = f \circ g$ is concave up

Without it, it may fail: $f(u) = e^{-u}$ (concave up, decreasing),

$g(x) = x^2 \implies h(x) = e^{-x^2}$

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \implies$ first term ≥ 0; $g'' > 0$

$\implies h''(x) > 0$ whenever $f'(g(x)) > 0$

Condition: f increasing ($f' > 0$) $\implies$ $h = f \circ g$ is concave up

Without it, it may fail: $f(u) = e^{-u}$ (concave up, decreasing),

$g(x) = x^2 \implies h(x) = e^{-x^2}$

$$h''(x) = (4x^2 - 2)e^{-x^2}, \quad h''(0) = -2 < 0$$

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \implies$ first term ≥ 0; $g'' > 0$

$\implies h''(x) > 0$ whenever $f'(g(x)) > 0$

Condition: f increasing $(f' > 0) \implies h = f \circ g$ is concave up

Without it, it may fail: $f(u) = e^{-u}$ (concave up, decreasing),

$g(x) = x^2 \implies h(x) = e^{-x^2}$

$$h''(x) = (4x^2 - 2)e^{-x^2}, \quad h''(0) = -2 < 0$$

$\implies h$ is concave down near 0

Challenging Problems 5

Suppose f and g are both concave up on $(-\infty, \infty)$. Under what condition on f will the composite function $h(x) = f(g(x))$ be concave up?

Solution

(Assume f, g twice differentiable.)

$$h'(x) = f'(g(x))\, g'(x)$$

$$h''(x) = f''(g(x))\, [g'(x)]^2 + f'(g(x))\, g''(x)$$

$f'' > 0$, $[g']^2 \geq 0 \implies$ first term ≥ 0; $g'' > 0$

$\implies h''(x) > 0$ whenever $f'(g(x)) > 0$

Condition: f increasing ($f' > 0$) $\implies$ $h = f \circ g$ is concave up

Without it, it may fail: $f(u) = e^{-u}$ (concave up, decreasing),

$g(x) = x^2 \implies h(x) = e^{-x^2}$

$$h''(x) = (4x^2 - 2)e^{-x^2}, \quad h''(0) = -2 < 0$$

$\implies h$ is concave down near 0

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution f is quadratic $\implies a \neq 0$

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution

$$f \text{ is quadratic} \implies a \neq 0$$
$$f'(x) = 2ax + b$$

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution

f is quadratic $\implies a \neq 0$

$f'(x) = 2ax + b$

$f''(x) = 2a \neq 0$ for all x

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution

f is quadratic $\implies a \neq 0$

$f'(x) = 2ax + b$

$f''(x) = 2a \neq 0$ for all x

f'' exists everywhere and is never 0:

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution

f is quadratic $\implies a \neq 0$

$f'(x) = 2ax + b$

$f''(x) = 2a \neq 0$ for all x

f'' exists everywhere and is never 0:
by Theorem 4.2.2 there is no candidate c with $f''(c) = 0$ or $f''(c)$ undefined

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution

f is quadratic $\implies a \neq 0$

$f'(x) = 2ax + b$

$f''(x) = 2a \neq 0$ for all x

f'' exists everywhere and is never 0:

by Theorem 4.2.2 there is no candidate c with $f''(c) = 0$ or $f''(c)$ undefined

($a > 0$: concave up on $\mathbb{R}$; $a < 0$: concave down on $\mathbb{R}$)

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution

f is quadratic $\implies a \neq 0$

$f'(x) = 2ax + b$

$f''(x) = 2a \neq 0$ for all x

f'' exists everywhere and is never 0:

by Theorem 4.2.2 there is no candidate c with $f''(c) = 0$ or $f''(c)$ undefined

($a > 0$: concave up on $\mathbb{R}$; $a < 0$: concave down on $\mathbb{R}$)

$\implies$ **f has no inflection points** ■

Communicate with Mathematics 2

Prove that a quadratic function $f(x) = ax^2 + bx + c$ has no inflection points.

Solution

f is quadratic $\implies a \neq 0$

$f'(x) = 2ax + b$

$f''(x) = 2a \neq 0$ for all x

f'' exists everywhere and is never 0:

by Theorem 4.2.2 there is no candidate c with $f''(c) = 0$ or $f''(c)$ undefined

($a > 0$: concave up on $\mathbb{R}$; $a < 0$: concave down on $\mathbb{R}$)

$\implies f$ **has no inflection points** ■

Thank you

Questions & discussion

Dr. Wessam Gouda · Section 4.2 — Concavity and Inflection Points